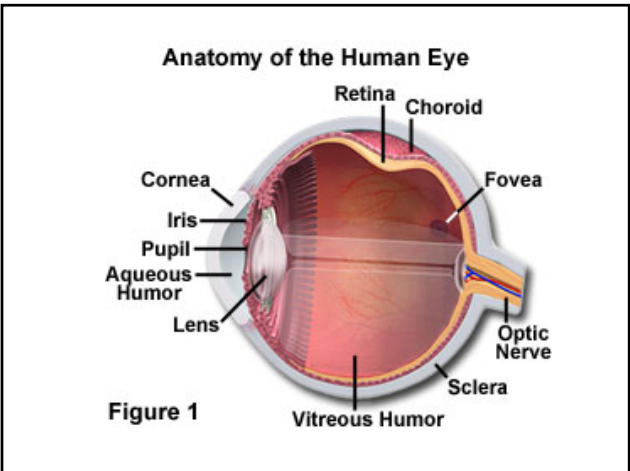
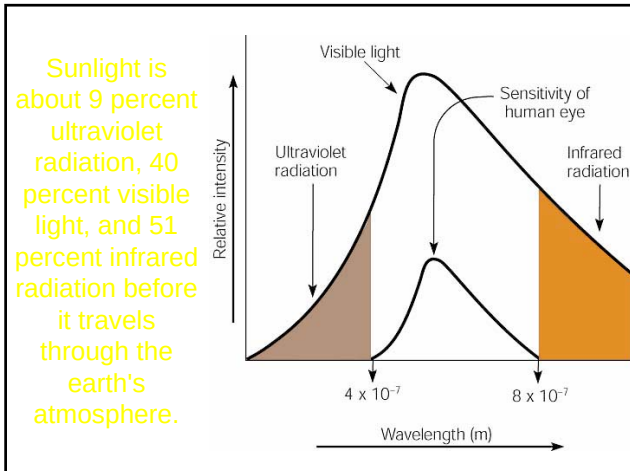
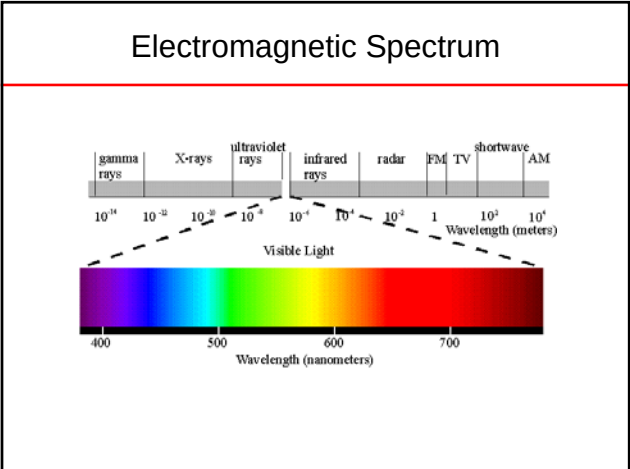


Human Visual System

From retina to perception
Image Analysis course
D.P. Huijsmans
16 feb 2006
With thanks to: Olympus, Michael Bach, psychology dept



Wavelength and accommodation dependent focus

Human Eye Accommodation Range

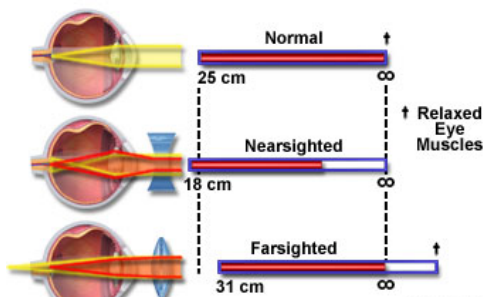


Figure 8

Distribution of rods and cones

Cone and Rod Cell Distribution in the Retina

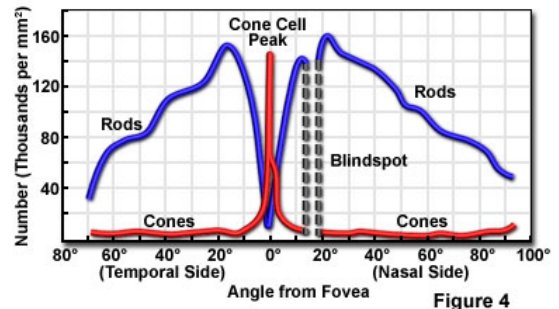


Figure 4

Cone detection profiles

Absorption Spectra of Human Visual Pigments

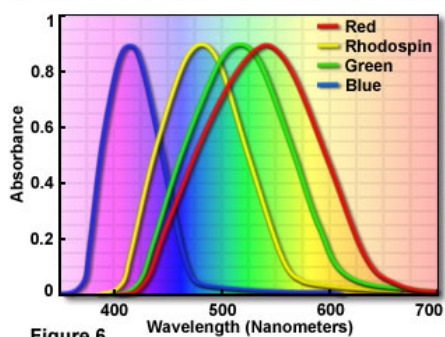


Figure 6

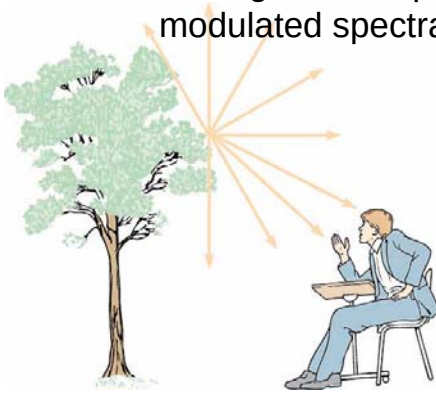
Detected light intensity: a game for 4 players



Light source: $E(\lambda)$
 Reflecting surface: $S(\lambda)$
 Absorbing medium: $A(\lambda)$
 Light from object:
 $A(\lambda) \cdot S(\lambda) \cdot E(\lambda)$
 Detector absorbance: $D(\lambda)$
 Intensity from object in direction detected:
 $= \int D(\lambda) \cdot A(\lambda) \cdot S(\lambda) \cdot E(\lambda) d(\lambda)$

The resulting intensity is direction and time dependent

Detecting and sampling modulated spectra

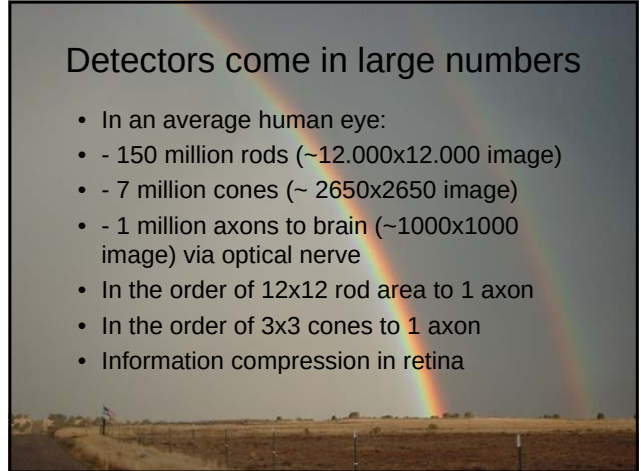


Directional spectral distribution is characterized by 4 numbers: I, R, G, B

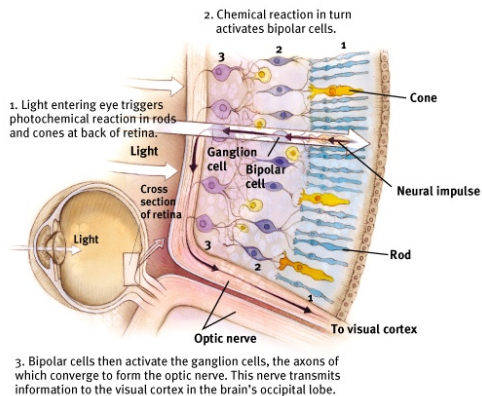
Physical spectrum Represented by only 4 numbers!

Detectors come in large numbers

- In an average human eye:
- - 150 million rods (~12.000x12.000 image)
- - 7 million cones (~ 2650x2650 image)
- - 1 million axons to brain (~1000x1000 image) via optical nerve
- In the order of 12x12 rod area to 1 axon
- In the order of 3x3 cones to 1 axon
- Information compression in retina

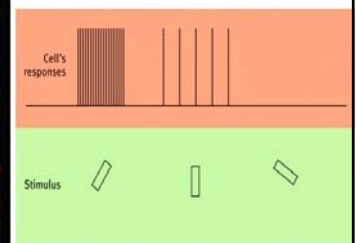


Upside down detection



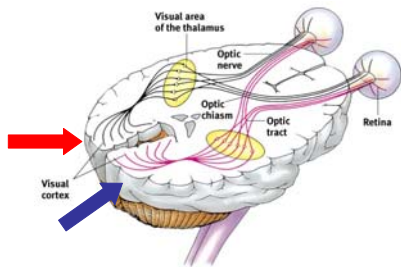
Retinal preprocessing:

Not only averaging also feature detection
(gradients, movements)



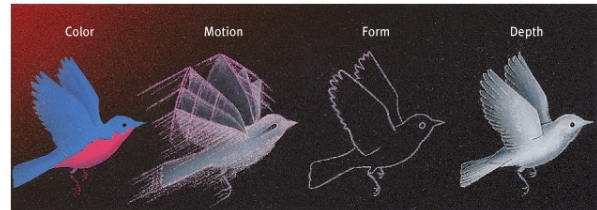
How visual signals travel in the brain
Visual Processing

Signals from the right side of each retina are sent to the left visual cortex

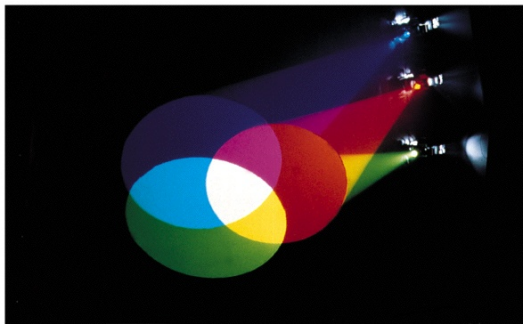


Human Visual System:

A highly parallel sense

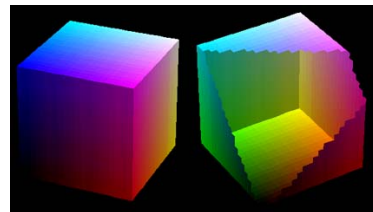
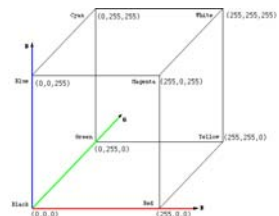


Light emission by RGB phosphor dots:

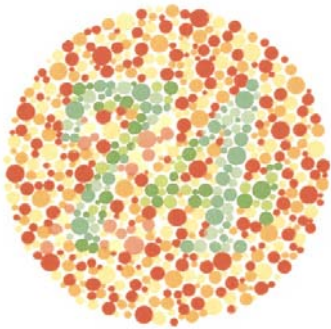


Additive color mixing

RGB color space
 at retinal level



Color blindness:
vision restricted to diagonal (rod intensity)
and 1 side plane of RGB-space (2 cones)



Perceived colors: opponent color space

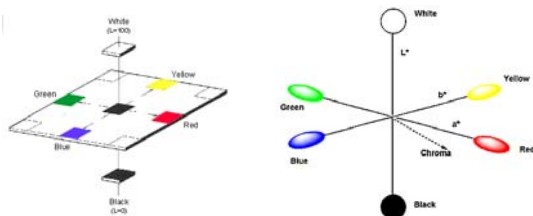
- RGB signals are combined before perception into sum and balance signals
- Red + Green = Yellow
- $\text{Max}\{\text{Red}, \text{Green}\} = \text{Redness xor Greenness}$
- $\text{Max}\{\text{Yellow}, \text{Blue}\} = \text{Yellowness xor Blueness}$

Color mixtures: positive combinations of:

- Redness and Yellowness : Hue quadrant around orange
- Redness and Blueness: Hue quadrant around Magenta
- Greenness and Yellowness: Hue quadrant
- Greenness and Blueness: Hue quadrant around Cyan

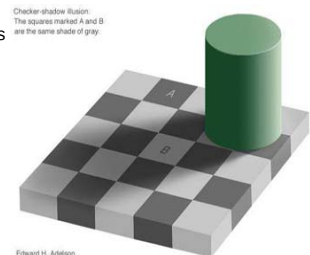


Opponent color: CIE Lab

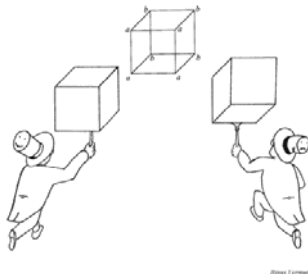


Optical illusions

- Simultaneous contrast
- Gradient addition: Mach bands
- Adaptation to flow field
- Induced motion
- After effects ; opponent color
- Foreground-background ambiguity : multiple solutions
- Velocity from peripheral vision
- Best site: Michael Bach: optical illusions



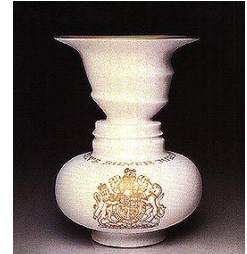
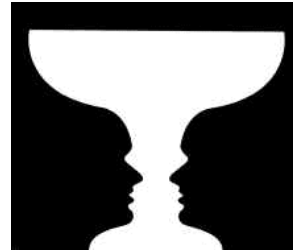
Wittgenstein: what is the logic of color?



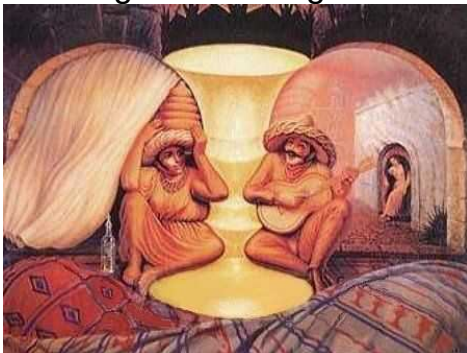
Die Welt des Glücklichen ist eine andere als die des Unglücklichen
(Stelling 6.43 van de Tractatus logico-philosophicus van Ludwig Wittgenstein)

Optical illusion: foreground-background

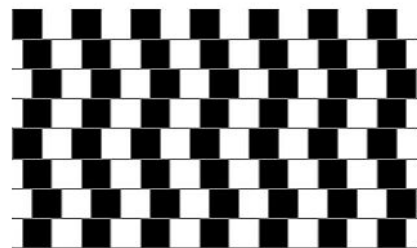
- Elizabeth-Philip vase



Optical illusion: extreme case foreground background

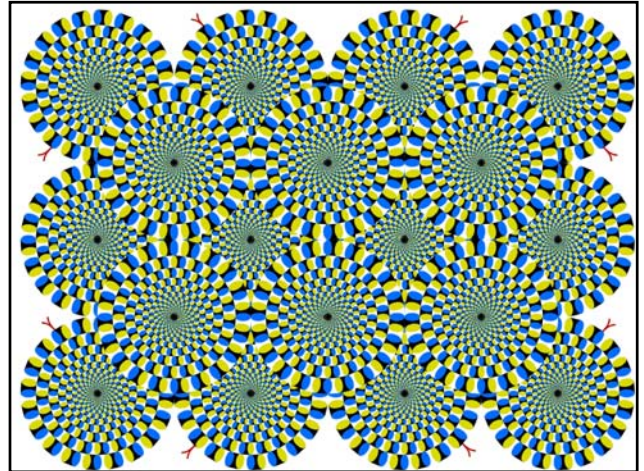
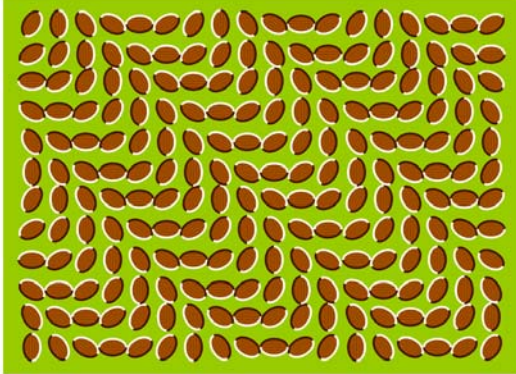


Optical illusion: parallel lines?



Are the horizontal lines parallel or do they slope?

Optical illusion: induced motion



Color Spaces for perception, design, display, print

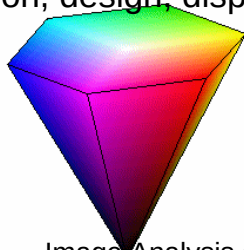
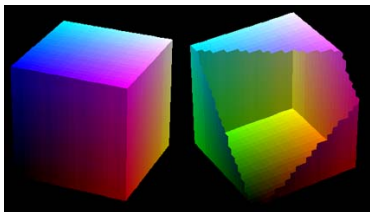
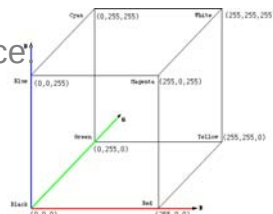


Image Analysis course
D.P. Huijsmans 16 feb 2006
With thanks to Internet

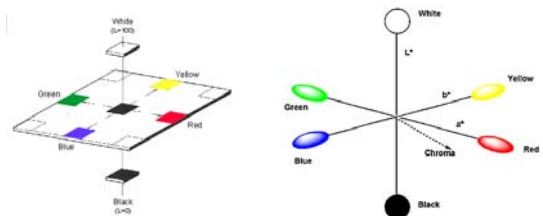
Color Spaces

- Human detection: RGB
- Human perception: opponent color
- Design: HSV, HSL
- Monitor CRT: additive system
- Printing: subtractive system CMY, CMYK
- False color display
- Color labeled visualizations

Detection color space
RGB color space
at retinal level

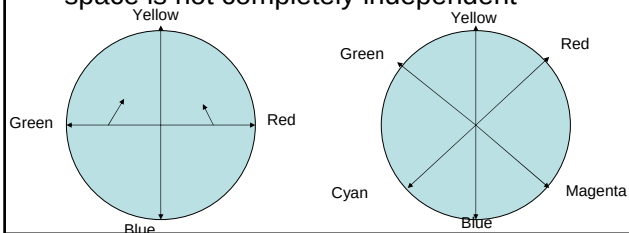


Perception color space
Opponent color: CIE Lab (Luv)

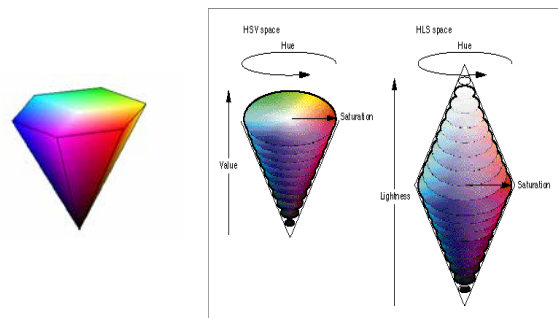


Design color space Distorted opponent color: HSV/HSL

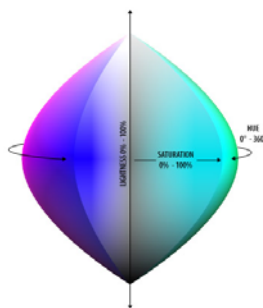
- Because Red and Green are associated with both Yellow (+) and red-green balance, the perceived opponent color space is not completely independent



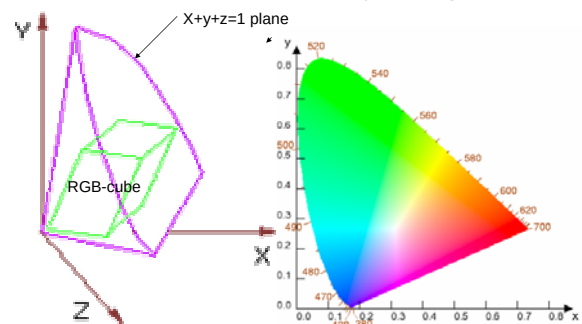
HSV cone: HSL double cone



HSL: better fitted to monitor near black and near white no color

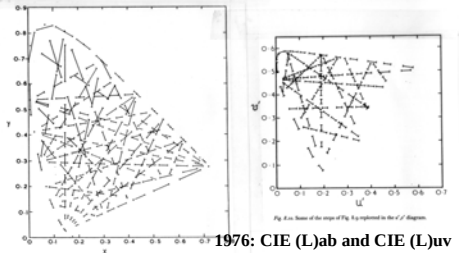


Design color: only Hue + Saturation CIE XYZ: Chromaticity Diagram



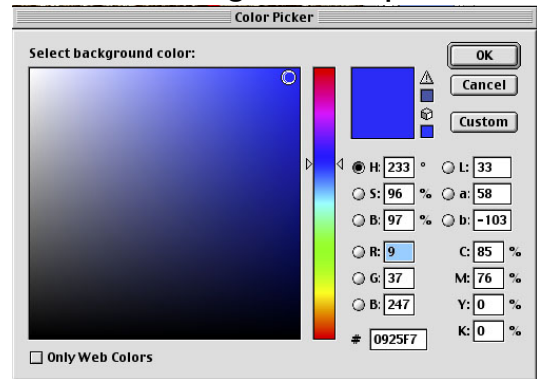
Color Measurement: CIE

- 1931: CIE XY(Z)
- Perceptually non-uniform color space:
- Small regions near blue, large near green



Perceptually uniform color spaces
Areas around colors for smallest noticeable color change ~ equal
(from Hunt: Colour reproduction)

Color design: color picker



CMY color space: Filter/ink/paint that absorbs a primary color (RGB) from white light



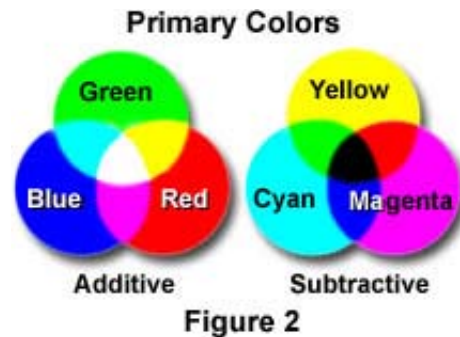
Subtractive color mixing

No filtering: white light
Cyan filter:
absorbs red light: green + blue left
Magenta filter:
Absorbs green light: blue + red left
Yellow filter:
Absorbs blue light: red + green left

C + M filters: blue left
M + Y filters: red left
C + Y filters: green left

C + M + Y filters: black, no light left

Looking from opposite sides along the diagonal of the color cube



Gamma-Correction

- Non-linear transfer of display intensity:
 - low intensities -> lighter
 - high intensities -> not well discernible
 - (Light perceived spreading against dark background)
- Non-linear transfer of printed intensity:
 - low intensities -> not well discernible
 - high intensities -> darker
 - (dark ink is perceived spreading against white background)
- Correction with mirrored (along diagonal linear ramp) function
- Printed results can not be judged from monitor:
- Calibration functions needed between scan, display, print

Displayed/printed pixel intensity

Stored pixel intensity

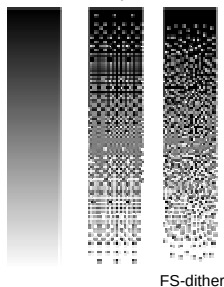
Color Calibration scanner, design, monitor, print



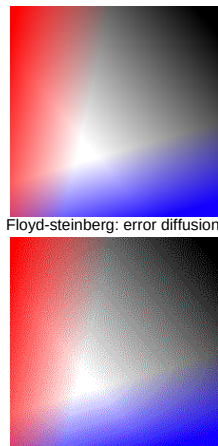
Color calibration card with ruler and gray scale

Dithered printing

- Gray or color values: pattern of uniform print dots



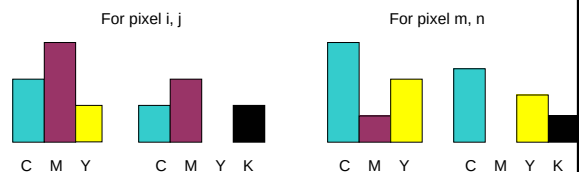
FS-dither



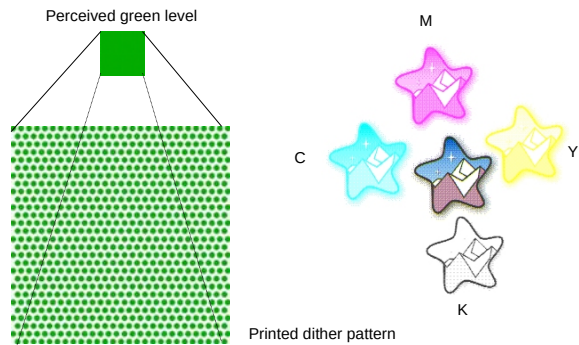
Floyd-steinberg: error diffusion

4-color printing: CMY->CMYK

- Gray values badly represented by CMY
- Extra black ink: (up to)common CMY fraction
- Per pixel: 2 colors + 1 black dither pattern
- Per image: four color print
- Saves (more expensive) color inks



CMYK printing and perception



Color printing: both additive and subtractive

- Mixed inks would be completely subtractive and have

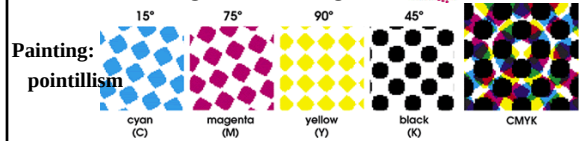
triangular gamut

- Rotated dither screens:

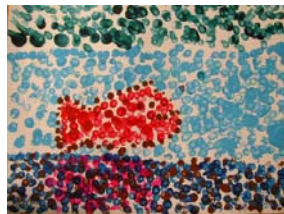
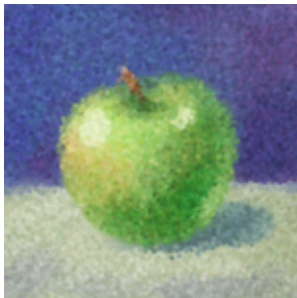
partly additive

- Result: printer gamut

more hexagonal then triangular



Pointillism: Seurat



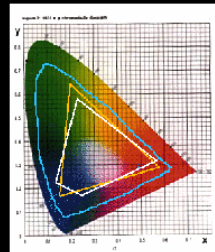
Paul Seurat



Perceivable, displayable and printable colors: gamuts

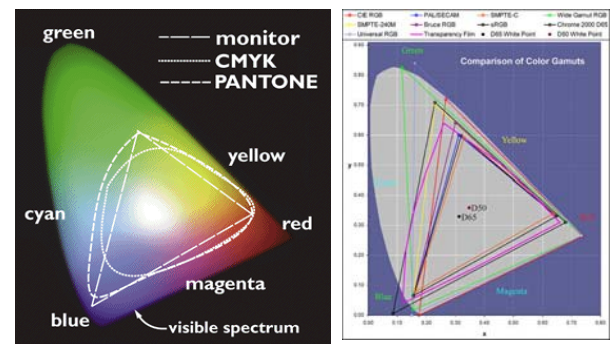
CIE 1931 xy chromaticity diagram

Commission Internationale de l'Éclairage



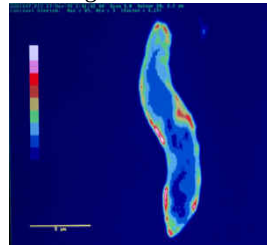
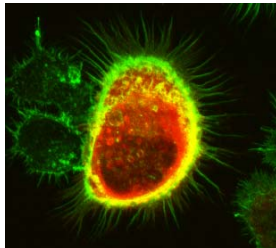
Film
Monitor
Printing Press

Gamut mapping

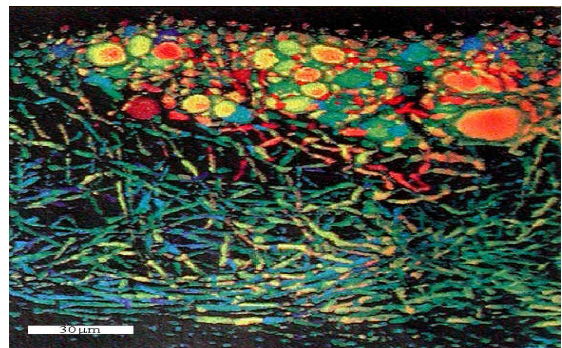


False/pseudo color display

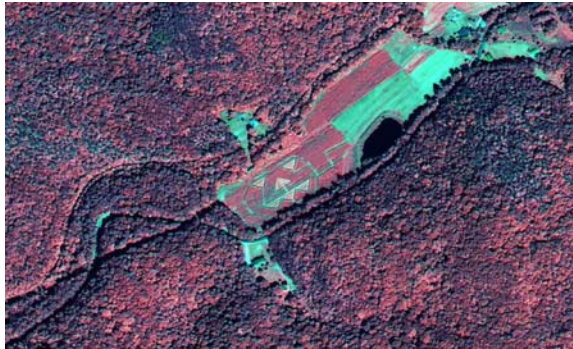
- Gray values via CLUT (color scale):
- Scale needed to understand color segmentation



False color changes only suggests tissue changes



Pseudo-Color visualization of differently labeled crops: true color coherence



Visualization of reconstructed zebra fish: color labeled tissues

